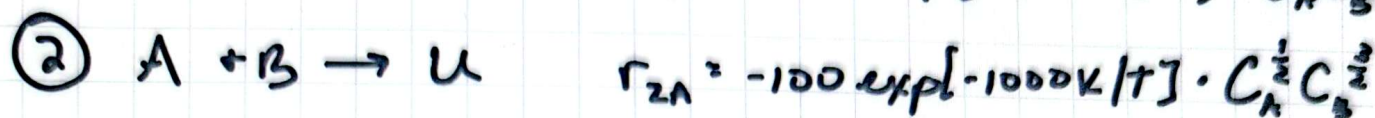
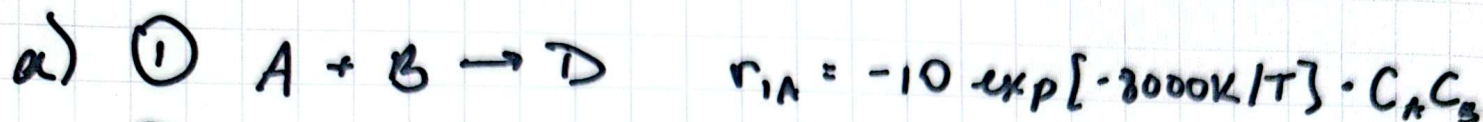


P 8-7

$$S = \frac{k_D}{k_U} C_A^{(\alpha_1 - \alpha_2)} C_B^{(\beta_1 - \beta_2)}$$

$$\left. \begin{aligned} k_D &= -10 \exp[-8000K/T] \\ k_U &= -100 \exp[-10000K/T] \end{aligned} \right\} \frac{k_D}{k_U} = \frac{1}{10} \exp\left[-\frac{2000K}{T}\right]$$

$E_D > E_U \therefore \text{high } T \uparrow S$

$\alpha_1 > \alpha_2 \therefore \text{high } C_A \uparrow S$

$\beta_1 < \beta_2 \therefore \text{low } C_B \uparrow S$

Based on the nature of these rxns, a semi-batch reactor that slowly adds B into A should be used & operated @ high temperatures.

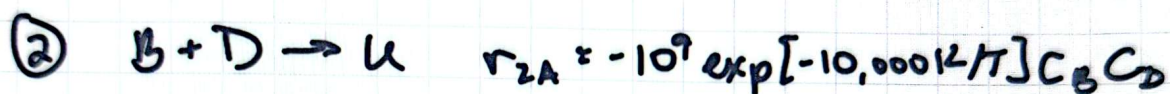
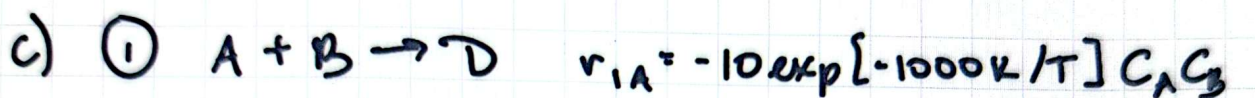


$$\frac{k_D}{k_U} = 10^{-4} \exp\left[\frac{7000K}{T}\right]$$

$E_D < E_U \therefore \text{low } T \uparrow S$

$\alpha_1 = \alpha_2 \quad \beta_1 = \beta_2$

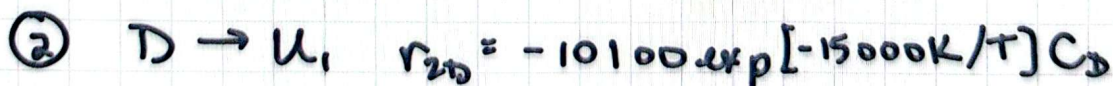
The reactor should be kept @ lower temperatures to prevent the production of undesired products. A PFR may be a good choice of reactor to maximize conversion while minimizing volume.



$$\frac{k_D}{k_U} = 10^{-9} \exp\left[\frac{9000K}{T}\right]$$

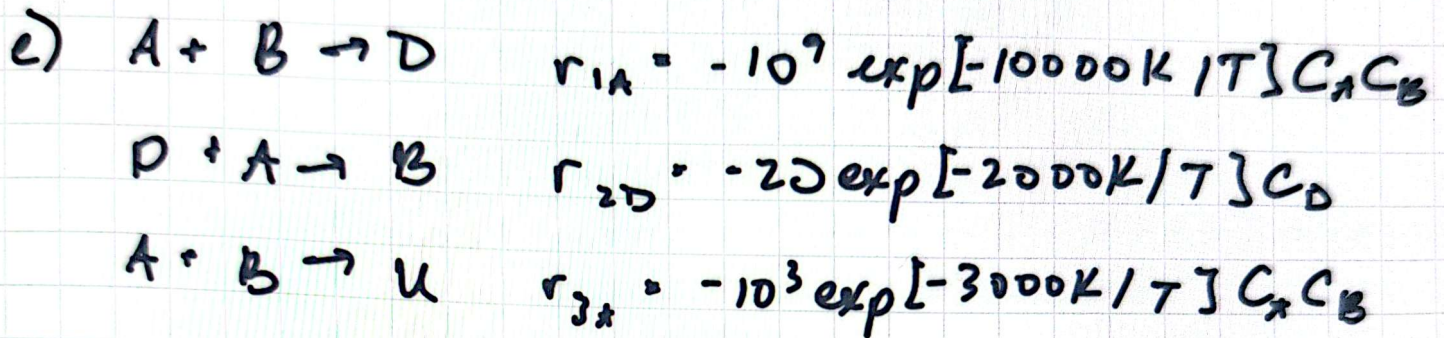
$$E_D < E_U \therefore \text{low } T \uparrow S$$

The reactor should be kept @ lower temperatures to prevent the undesired rxn w/ the higher activation energy. If possible, a membrane PFR should be used to remove product D before it can react w/ B. If a membrane is not possible, a CSTR w/ high concentrations of A and low of B should be used.

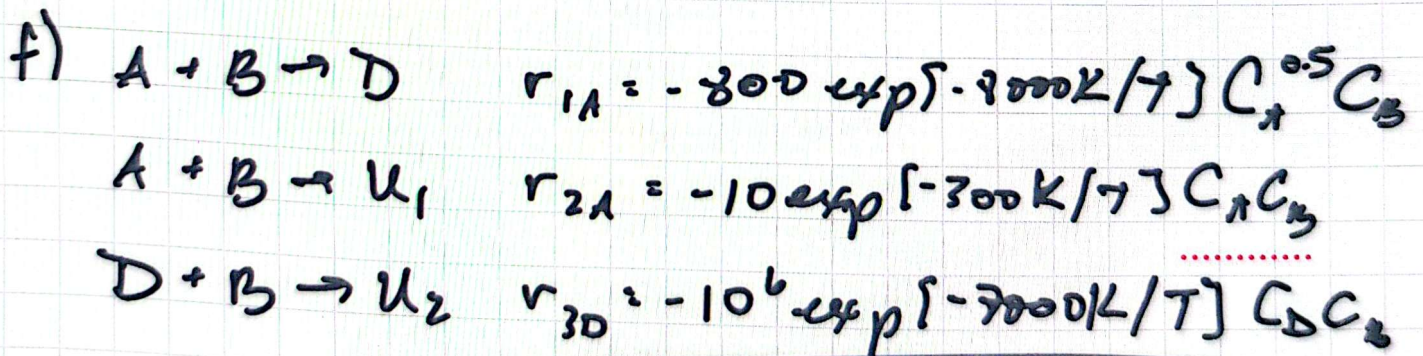


$$E_D < E_{U_1} < E_{U_2} \therefore \text{low } T \uparrow S$$

The rxr should be kept @ lower temperatures to avoid the undesired rxn's w/ higher activation energies. The best rxr design would be a membrane rxr that removes D.



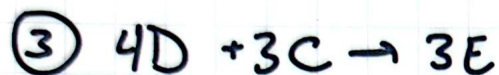
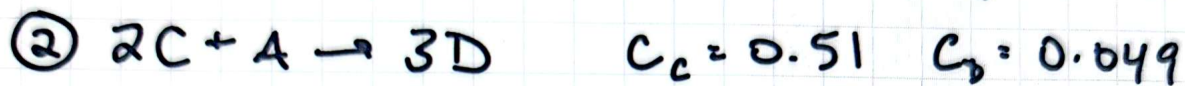
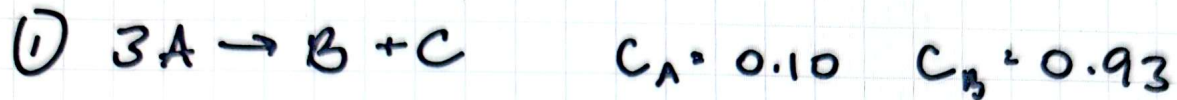
Best rxr design = semi-batch or membrane rxr, add A slowly / remove D as it's formed. Keep @ high temp.



Keep A @ low concentrations, remove D as formed.

P 8-11

liquid phase rxn's, CSTR @ $T = 325 \text{ K}$



$$r_{1A} = -7.0(0.1) = -0.7 \text{ mol/dm}^3 \cdot \text{min}$$

$$r_{2D} = 3.0(0.51)^2(0.1) = 0.07803 \text{ mol/dm}^3 \cdot \text{min}$$

$$r_{3E} = 2.0(0.049)(0.51) = 0.04998 \text{ mol/dm}^3 \cdot \text{min}$$

1) $r_{1A} = -0.7 \text{ mol/dm}^3 \cdot \text{min}$

$$r_{3A} = 0$$

$$r_{2A} = -\frac{1}{3}r_{2D} = -\frac{1}{3}(0.07803)$$

$r_{1A} = -0.7 \text{ mol/dm}^3 \cdot \text{min}$
$r_{2A} = -0.02601 \text{ mol/dm}^3 \cdot \text{min}$
$r_{3A} = 0 \text{ mol/dm}^3 \cdot \text{min}$

2) $r_{1B} = -\frac{1}{3}r_{1A} = -\frac{1}{3}(-0.7)$

$r_{1B} = 0.233 \text{ mol/dm}^3 \cdot \text{min}$
$r_{2B} = r_{3B} = 0$

$$3) \quad r_{1C} = -\frac{1}{3} r_{1A} = -\frac{1}{3}(-0.7)$$

$$r_{2C} = -\frac{2}{3} r_{2D} = -\frac{2}{3}(0.07803)$$

$$r_{3C} = -r_{3E}$$

$$r_{1C} = 0.233 \text{ mol/dm}^3 \cdot \text{min}$$

$$r_{2C} = -0.052 \text{ mol/dm}^3 \cdot \text{min}$$

$$r_{3C} = -0.04998 \text{ mol/dm}^3 \cdot \text{min}$$

$$4) \quad r_{1D} = 0$$

$$r_{2D} = r_{2D}$$

$$r_{3D} = -\frac{4}{3} r_{3E} = -\frac{4}{3}(0.04998)$$

$$r_{1D} = 0$$

$$r_{2D} = 0.07803 \text{ mol/dm}^3 \cdot \text{min}$$

$$r_{3D} = -0.06664 \text{ mol/dm}^3 \cdot \text{min}$$

$$5) \quad r_{1E} = r_{2E} = 0$$

$$r_{3E} = 0.04998 \text{ mol/dm}^3 \cdot \text{min}$$

$$6) \quad r_A = r_{1A} + r_{2A} + r_{3A}$$

$$r_B = r_{1B} + r_{2B} + r_{3B} \quad \dots \text{ etc}$$

$$r_A = -0.72601 \text{ mol/dm}^3 \cdot \text{min}$$

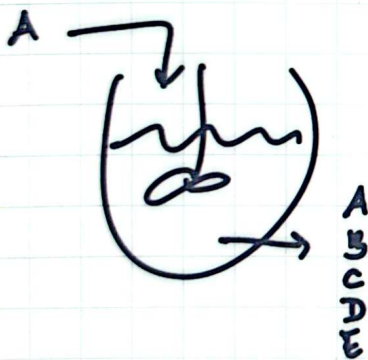
$$r_B = 0.233 \text{ mol/dm}^3 \cdot \text{min}$$

$$r_C = 0.13102 \text{ mol/dm}^3 \cdot \text{min}$$

$$r_D = 0.101139 \text{ mol/dm}^3 \cdot \text{min}$$

$$r_E = 0.04998 \text{ mol/dm}^3 \cdot \text{min}$$

$$7) v_0 = 100 \text{ dm}^3/\text{min} \quad C_{A0} = 3 \text{ mol/dm}^3$$



$$F_{A0} = v_0 C_{A0} = 300 \text{ mol A/min}$$

$$0 = F_{A0} - F_A + r_A V$$

$$F_A = v_0 C_A = 100(0.1) = 10 \text{ mol/min}$$

$$0 = 300 - 10 - 0.7V$$

$$\boxed{V = 414 \text{ dm}^3}$$

$$8) F_A = v_0 C_A = 100(0.1) = 10 \text{ mol/min}$$

$$F_B = 100(0.93) = 93 \text{ mol/min}$$

$$F_C = 51 \text{ mol/min} \quad F_D = 4.9 \text{ mol/min}$$